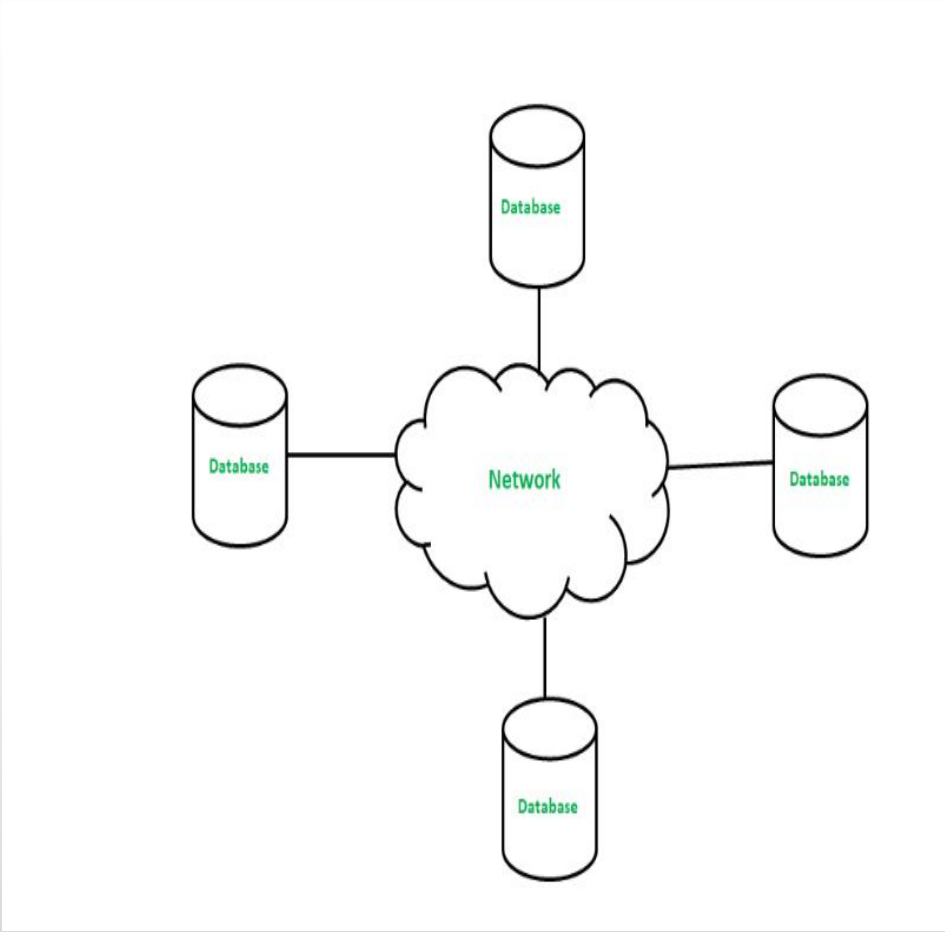
Distributed DB
&
Data Warehouse

Distributed DB

- ✔ A distributed database is basically a database, it is spread over different sites, i.e, on multiple computers or over a network of computers.
- ✔ A distributed database system is located on various sites that don't share physical components.
- ✔ This may be required when a particular database needs to be accessed by various users globally.



Types(Distributed DB):

- **Homogeneous Database**:
  In a homogeneous database, all different sites store database identically.
- The operating system, database management system, and the data structures used – all are the same at all sites. Hence, they’re easy to manage.

Types(Distributed DB):

- **2. Heterogeneous Database :**
  In a heterogeneous distributed database, different sites can use different schema and software that can lead to problems in query processing and transactions.
- Different computers may use a different operating system, different database application. They may even use different data models for the database.
- Hence, translations are required for different sites to communicate.

Distributed Data Storage

- **Method 1: Replication**
- In this approach, the entire relationship is stored redundantly at 2 or more sites. If the entire database is available at all sites, it is a fully redundant database. Hence, in replication, systems maintain copies of data.

Distributed Data Storage

Replication:

Advantages:

✔ Increases the availability of data

✔ Query requests can be processed in parallel.

Disadvantage:

✔ Data needs to be constantly updated

✔ concurrency control becomes way more complex

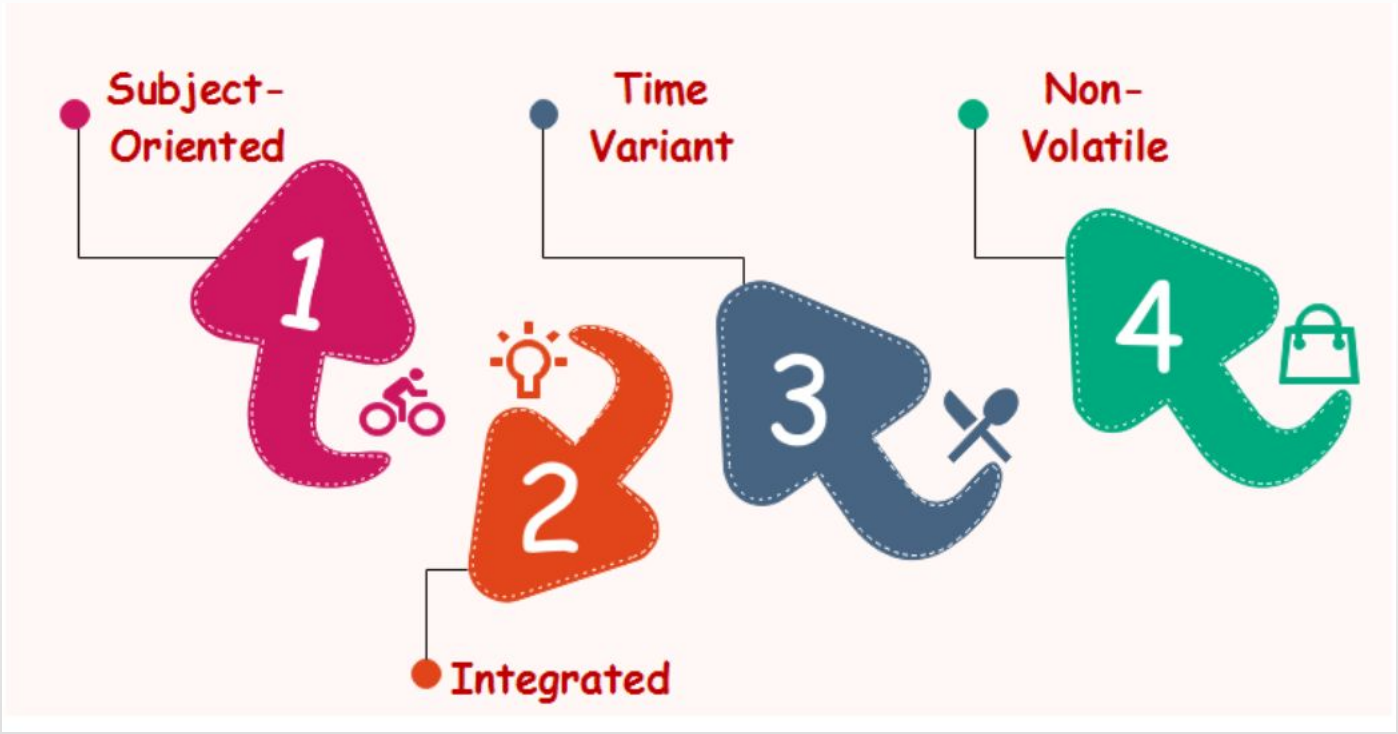
Distributed Data Storage

- **Method 2: Fragmentation**
- The relations are fragmented (i.e., they're divided into smaller parts) and each of the fragments is stored in different sites where they're required.
- It must be made sure that the fragments are such that they can be used to reconstruct the original relation (i.e, there isn't any loss of data).

Data warehouse

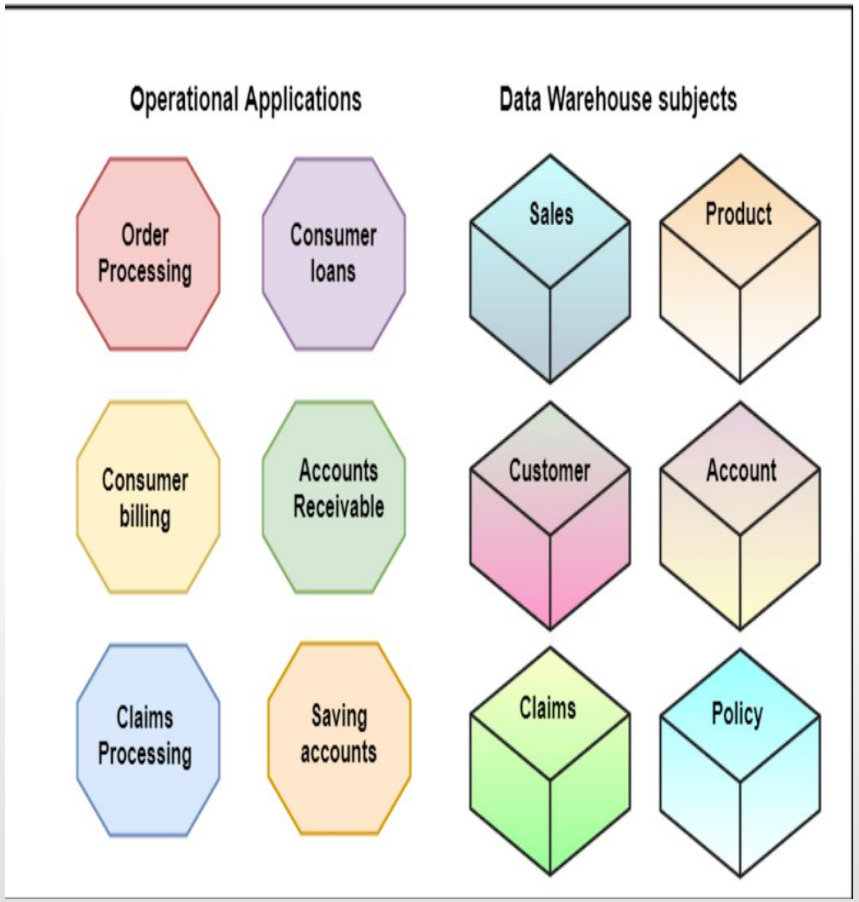
- ✔ Data Warehouse (DW) is a relational database that is designed for query and analysis rather than transaction processing.
- ✔ It includes historical data derived from transaction data from single and multiple sources.
- ✔ Data Warehouse provides integrated, enterprise-wide, historical data and focuses on providing support for decision-makers for data modeling and analysis.

Key features of Datawarehouse



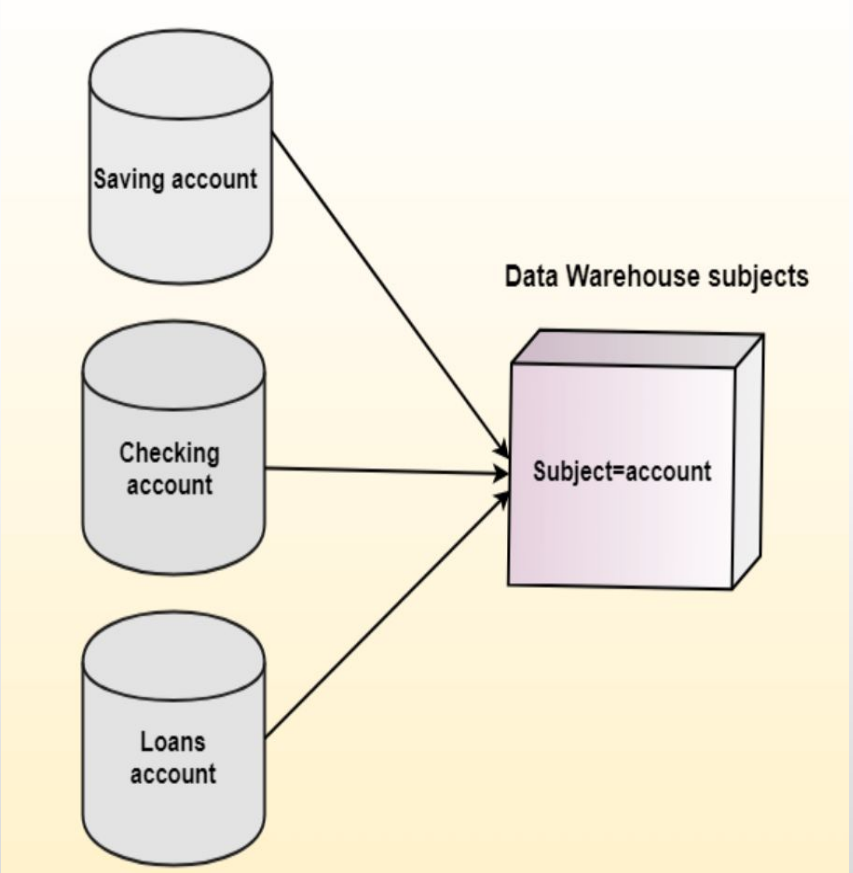
Subject Oriented

- A data warehouse target on the modeling and analysis of data for decision-makers.
- It provides a concise and straightforward view around a particular subject, such as customer, product, or sales, instead of the global organization's ongoing operations.



Integrated

- A data warehouse integrates various heterogeneous data sources like RDBMS, flat files, and online transaction records.
- It requires performing data cleaning and integration during data warehousing to ensure consistency in naming conventions, attributes types, among different data source



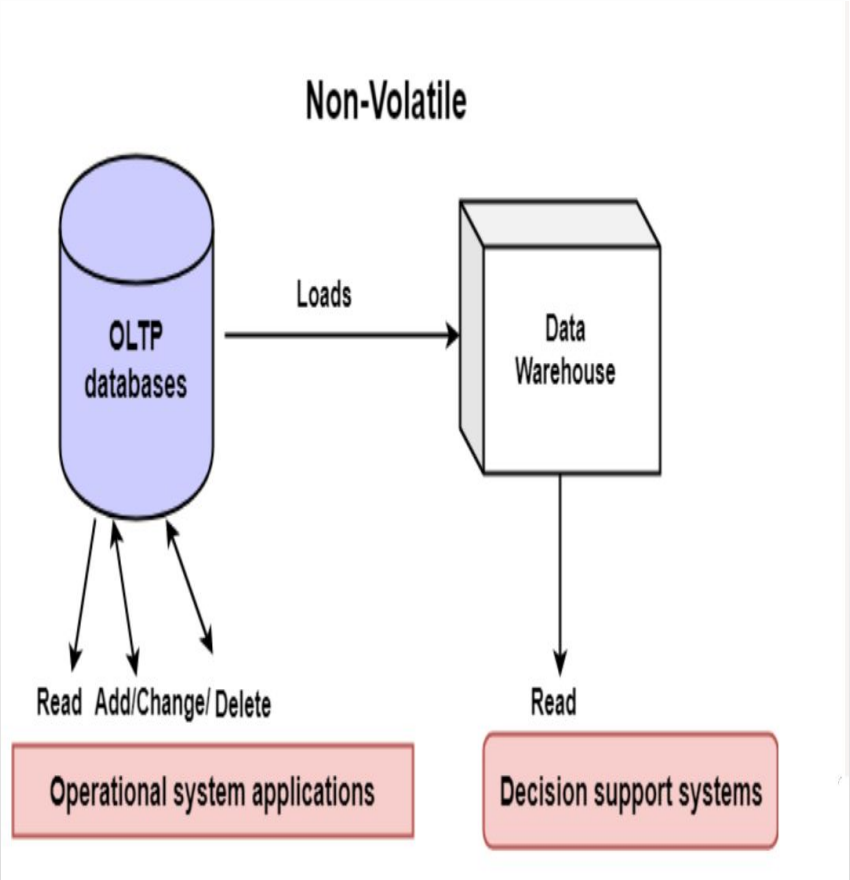
Time Variant

- Historical information is kept in a data warehouse. For example, one can retrieve files from 3 months, 6 months, 12 months, or even previous data from a data warehouse.



Non Volatile

- The operational updates of data do not occur in the data warehouse, i.e., update, insert, and delete operations are not performed.
- It usually requires only two procedures in data accessing: Initial loading of data and access to data.



Goals of Data Warehousing

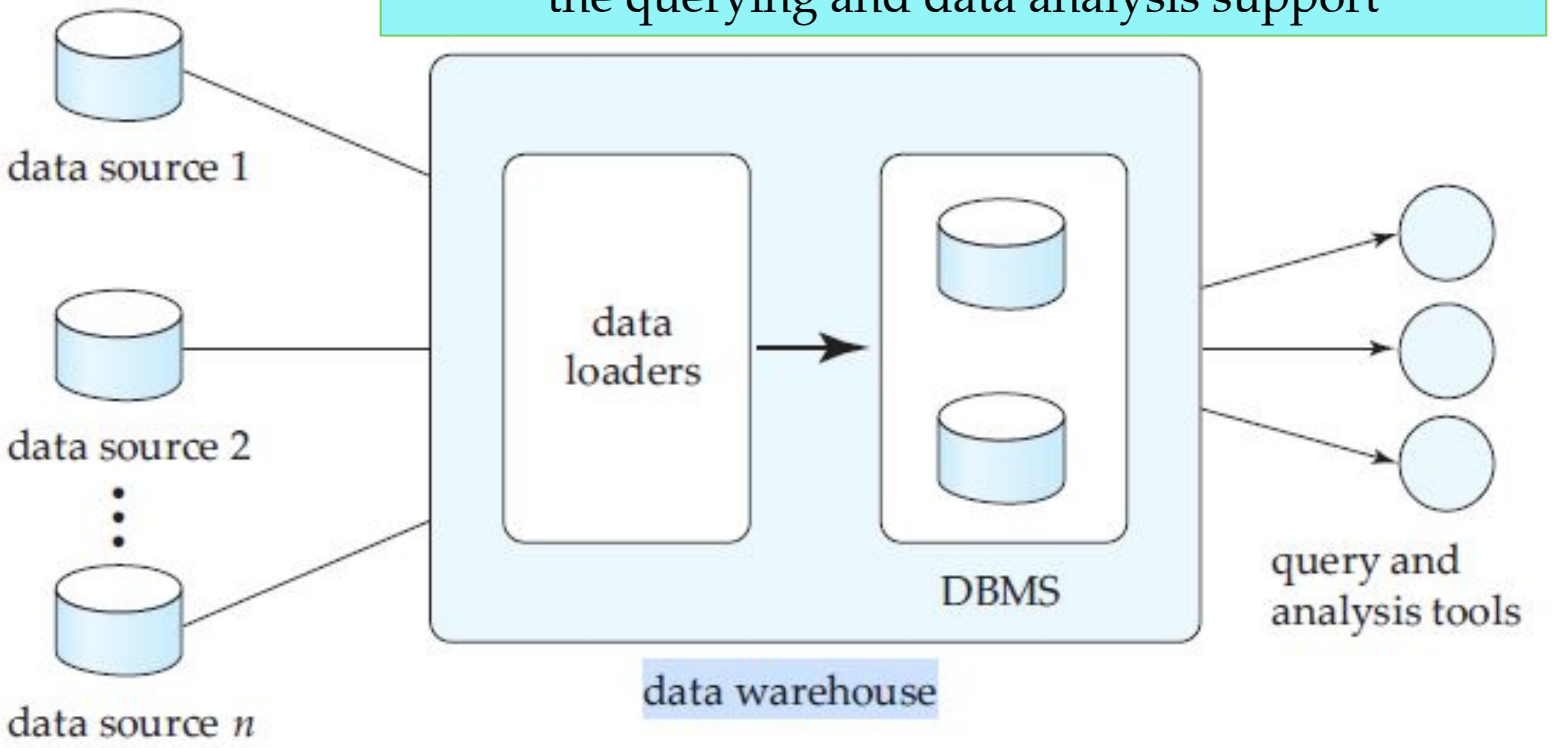
- To help reporting as well as analysis
- Maintain the organization's historical information
- Be the foundation for decision making.

Need of Data warehouse



Data Warehouse architecture

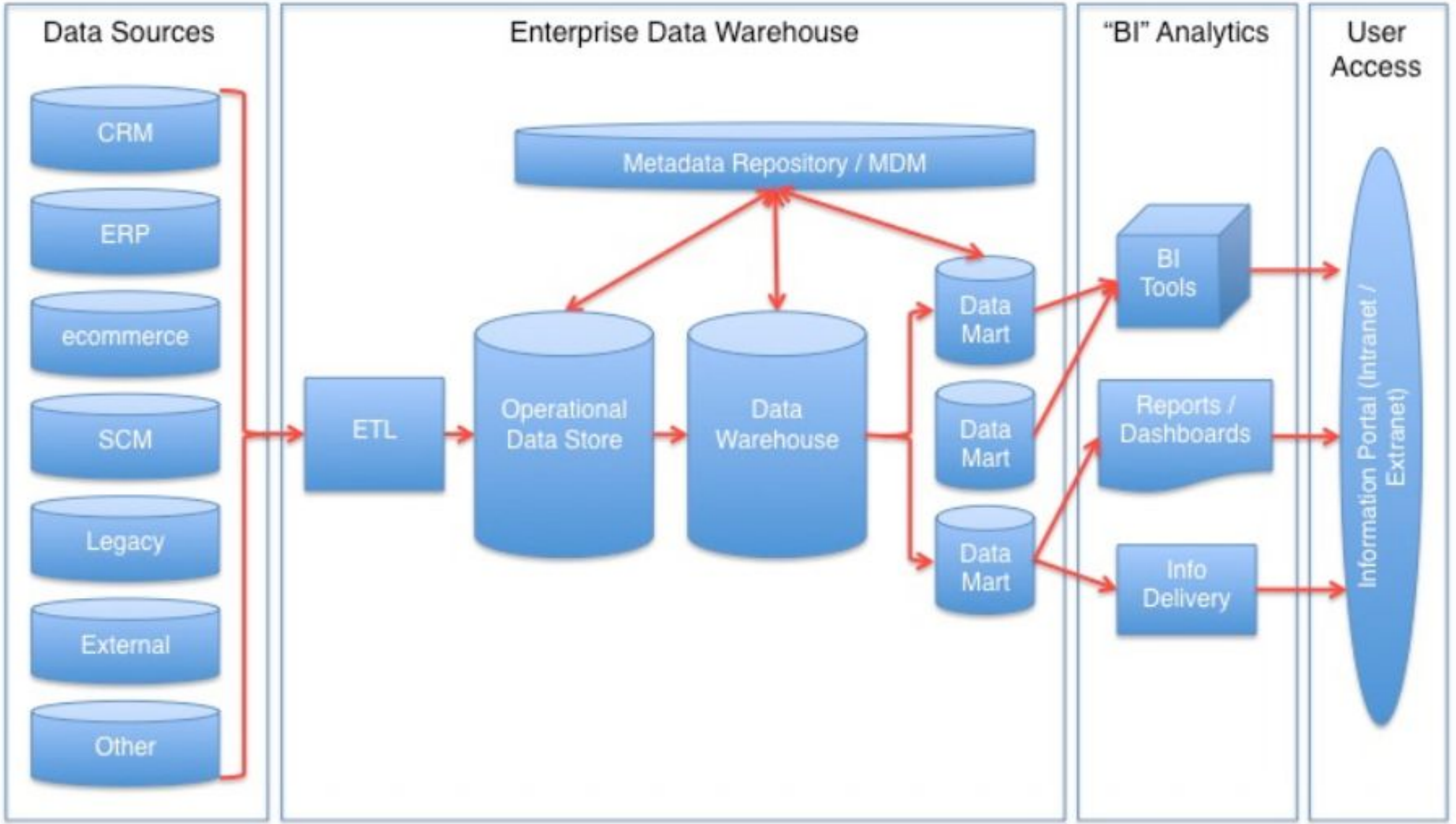
Illustrates gathering of data, the storage of data, and the querying and data analysis support



The different steps involved in getting data into a data warehouse are called **extract, transform, and load** or **ETL** tasks;

Extraction refers to getting data from the sources,
Loading refers to loading the data into the data warehouse

A


Data Sources
Enterprise Data Warehouse
"BI" Analytics
User Access
CRM
ERP
ecommerce
SCM
Legacy
External
Other
Metadata Repository / MDM
ETL
Operational Data Store
Data Warehouse
Data Mart
Data Mart
Data Mart
BI Tools
Reports / Dashboards
Info Delivery
Information Portal (Intranet / Extranet)

Data warehouse components

Data transformation and cleansing.

- The task of correcting and preprocessing data is called data cleansing.
- Data sources often deliver data with numerous minor inconsistencies, which can be corrected.
- Ex : Names are often misspelled, Addresses may have street, area, or city names misspelled, or postal codes entered incorrectly.

How to propagate updates.

- Updates on relations at the data sources must be propagated to the data warehouse.
- If the relations at the data warehouse are exactly the same as those at the data source, the propagation is straightforward.
- If they are not, the problem of propagating updates is basically the view-maintenance problem

Data Mart

- A Data Mart is a subset of a directorial information store, generally oriented to a specific purpose or primary data subject which may be distributed to provide business needs. Data Marts are analytical record stores designed to focus on particular business functions for a specific community within an organization.

Reasons for creating a data mart

- Easy access to frequently needed data
- Creates collective view by a group of users
- Improves end-user response time
- Ease of creation in less time
- Lower cost than implementing a full Data warehouse
- Potential users are more clearly defined than in a full Data warehouse
- Simple data maintenance

DBS Vs Data Warehouse

Database System	Data Warehouse
It supports operational processes.	It supports analysis and performance reporting.
Capture and maintain the data.	Explore the data.
Current data.	Multiple years of history.
Data is balanced within the scope of this one system.	Data must be integrated and balanced from multiple system.